Area in Polar Coordinates

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: The polar area formula**

State the formula for the area enclosed by a polar curve $r = f(\theta)$ between $\theta = \alpha$ and $\theta = \beta$.

$A = \frac{1}{2}\int_{\alpha}^{\beta} r^2\,d\theta = \frac{1}{2}\int_{\alpha}^{\beta} [f(\theta)]^2\,d\theta.$

2 **[Foundational] • Target: The polar area formula**

Find the area enclosed by one full loop of $r = 3\cos\theta$ (i.e. $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$).

$A = \frac{1}{2}\int_{-\pi/2}^{\pi/2} 9\cos^2\theta\,d\theta = \frac{9}{2}\int_{-\pi/2}^{\pi/2} \frac{1+\cos 2\theta}{2}\,d\theta = \frac{9}{4}\left[\theta + \frac{\sin 2\theta}{2}\right]_{-\pi/2}^{\pi/2} = \frac{9}{4}(\pi) = \frac{9\pi}{4}.$

3 **[Foundational] • Target: Area of a full cardioid**

Find the total area enclosed by the cardioid $r = 1 + \cos\theta$.

$A = \frac{1}{2}\int_{0}^{2\pi} (1+\cos\theta)^2\,d\theta = \frac{1}{2}\int_{0}^{2\pi} (1 + 2\cos\theta + \cos^2\theta)d\theta$. Using $\cos^2\theta = \frac{1+\cos 2\theta}{2}$:
$= \frac{1}{2}\left[\theta + 2\sin\theta + \frac{\theta}{2} + \frac{\sin 2\theta}{4}\right]_0^{2\pi} = \frac{1}{2}(2\pi + \pi) = \frac{3\pi}{2}.$

4 **[Foundational] • Target: Area between two polar curves**

Find the area inside the circle $r = 3$ but outside the circle $r = 2$.

$A = \frac{1}{2}\int_{0}^{2\pi} (3^2 - 2^2)d\theta = \frac{1}{2}\int_{0}^{2\pi} 5\,d\theta = \frac{1}{2}(5)(2\pi) = 5\pi.$

5 **[Foundational] • Target: Area of one petal of a rose curve**

Find the area of one petal of $r = 2\sin(2\theta)$, using the interval $\theta \in [0, \frac{\pi}{2}]$ where one petal is traced.

$A = \frac{1}{2}\int_{0}^{\pi/2} 4\sin^2(2\theta)\,d\theta = 2\int_{0}^{\pi/2} \frac{1-\cos 4\theta}{2}\,d\theta = \left[\theta - \frac{\sin 4\theta}{4}\right]_0^{\pi/2} = \frac{\pi}{2}.$

6 **[Foundational] • Target: Setting up area between curves that intersect**

Set up (but do not evaluate) the integral for the area inside both $r = 2$ and $r = 2 + 2\cos\theta$, noting they intersect where $2 = 2 + 2\cos\theta$, i.e. $\theta = \pm\frac{\pi}{2}$.

By symmetry, for $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ the circle $r = 2$ is the inner (smaller) boundary, while for $\theta \in [\frac{\pi}{2}, \frac{3\pi}{2}]$ the cardioid is inner. Area $= \frac{1}{2}\int_{-\pi/2}^{\pi/2} 2^2\,d\theta + \frac{1}{2}\int_{\pi/2}^{3\pi/2} (2+2\cos\theta)^2\,d\theta.$

7 **[Foundational] • Target: Area of a full rose curve**

Find the total area enclosed by all 4 petals of $r = 3\cos(2\theta)$, using the fact that one petal is traced for $\theta \in [-\frac{\pi}{4}, \frac{\pi}{4}]$.

Area of one petal: $\frac{1}{2}\int_{-\pi/4}^{\pi/4} 9\cos^2(2\theta)\,d\theta = \frac{9}{2}\int_{-\pi/4}^{\pi/4} \frac{1+\cos 4\theta}{2}\,d\theta = \frac{9}{4}\left[\theta + \frac{\sin 4\theta}{4}\right]_{-\pi/4}^{\pi/4} = \frac{9}{4}\cdot\frac{\pi}{2} = \frac{9\pi}{8}$. Total for all 4 petals: $4 \times \frac{9\pi}{8} = \frac{9\pi}{2}$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Area inside a limaçon**

Find the area enclosed by the limaçon $r = 3 + \cos\theta$ (a curve without an inner loop, since the constant exceeds the coefficient).

$A = \frac{1}{2}\int_0^{2\pi}(3+\cos\theta)^2\,d\theta = \frac{1}{2}\int_0^{2\pi}(9+6\cos\theta+\cos^2\theta)d\theta = \frac{1}{2}\left[9\theta+6\sin\theta+\frac{\theta}{2}+\frac{\sin 2\theta}{4}\right]_0^{2\pi} = \frac{1}{2}(18\pi+\pi) = \frac{19\pi}{2}.$

9 **[Intermediate]** • **Target: Area inside a limaçon**

Find the area of the small inner loop of the limaçon $r = 1 + 2\cos\theta$ (constant less than coefficient, so an inner loop exists), traced for $\theta \in \left[\frac{2\pi}{3}, \frac{4\pi}{3}\right]$.

$A = \frac{1}{2}\int_{2\pi/3}^{4\pi/3}(1+2\cos\theta)^2\,d\theta = \frac{1}{2}\int_{2\pi/3}^{4\pi/3}(1+4\cos\theta+4\cos^2\theta)d\theta = \frac{1}{2}[3\theta+4\sin\theta+\sin 2\theta]_{2\pi/3}^{4\pi/3}$. At $\frac{4\pi}{3}$: $4\pi - 2\sqrt{3} + \frac{\sqrt{3}}{2}$. At $\frac{2\pi}{3}$: $2\pi + 2\sqrt{3} - \frac{\sqrt{3}}{2}$. Difference: $2\pi - 3\sqrt{3}$. $A = \frac{1}{2}\left(2\pi - 3\sqrt{3}\right) = \pi - \frac{3\sqrt{3}}{2} \approx 0.54$.

10 **[Intermediate]** • **Target: Setting up area problems (AP-style, no evaluation)**

Set up (but do not evaluate) the integral for the area inside the rose $r = 2\sin(3\theta)$ that lies in the first quadrant only, for one petal traced on $\theta \in \left[0, \frac{\pi}{3}\right]$.

$A = \frac{1}{2}\int_0^{\pi/3}[2\sin(3\theta)]^2\,d\theta = \frac{1}{2}\int_0^{\pi/3}4\sin^2(3\theta)\,d\theta.$

11 **[Intermediate]** • **Target: Setting up area problems (AP-style, no evaluation)**

Set up (but do not evaluate) the integral for the area between the circle $r = 5$ and the cardioid $r = 5 - 5\sin\theta$ (inside the circle, outside the cardioid), for $\theta \in [0, \pi]$.

Area $= \frac{1}{2}\int_0^{\pi}\left[5^2 - (5-5\sin\theta)^2\right]d\theta.$

12 **[Intermediate]** • **Target: Finding intersection points of polar curves**

Find all points of intersection of $r = 2\cos\theta$ and $r = 2\sin\theta$ (both circles of radius 1) by setting the equations equal.

$2\cos\theta = 2\sin\theta \Rightarrow \tan\theta = 1 \Rightarrow \theta = \frac{\pi}{4}$ or $\theta = \frac{5\pi}{4}$. At $\theta = \frac{\pi}{4}$: $r = 2\cos\left(\frac{\pi}{4}\right) = \sqrt{2}$. At $\theta = \frac{5\pi}{4}$: $r = 2\cos\left(\frac{5\pi}{4}\right) = -\sqrt{2}$ (a negative r, representing the same point reached differently). Both curves also pass through the pole ($r = 0$) at different angles, which must be checked separately.

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13 [Intermediate] • Target: Finding intersection points of polar curves

Explain why checking intersections of two polar curves algebraically (setting $f(\theta) = g(\theta)$) can MISS the pole as an intersection point, and why it must be checked separately.

Different curves can reach the pole ($r = 0$) at different values of θ (since a single point, the origin, corresponds to $r = 0$ for ANY θ). Setting $f(\theta) = g(\theta)$ only finds angles where both curves have the SAME r at the SAME θ, so it can miss cases where each curve passes through the origin at a different angle — the pole must always be checked as a separate case.

14 [Intermediate] • Target: Area using symmetry to simplify computation

Explain how the symmetry of $r = 3\cos(2\theta)$ about both axes allows the total area to be found by computing just one petal and multiplying, rather than integrating over the full range at once.

Since $r = 3\cos(2\theta)$ produces 4 congruent petals by symmetry (reflections across both the x- and y-axes map petals onto each other), computing the area of ONE petal via a single definite integral and multiplying by 4 avoids having to integrate piecewise across the full $[0, 2\pi]$ range while tracking sign changes.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Area review: full worked example • Cross-Lesson: 1.01 Integration by Parts

Find the area enclosed by $r = \sqrt{\theta}$ for $\theta \in [0, \pi]$ (a spiral arc closed off by the segment back to the pole).

$A = \frac{1}{2}\int_0^{\pi} (\sqrt{\theta})^2 \, d\theta = \frac{1}{2}\int_0^{\pi} \theta \, d\theta = \frac{1}{2}\left[\frac{\theta^2}{2}\right]_0^{\pi} = \frac{\pi^2}{4}$.

16 [Challenging] • Target: Area review: full worked example • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

Find the area enclosed by $r = 4\sin\theta$ (a circle) for $\theta \in [0, \pi]$, and verify your answer matches the known area formula for a circle of the appropriate radius.

$A = \frac{1}{2}\int_0^{\pi} 16\sin^2\theta \, d\theta = 8\int_0^{\pi} \frac{1 - \cos 2\theta}{2} \, d\theta = 4\left[\theta - \frac{\sin 2\theta}{2}\right]_0^{\pi} = 4\pi$. This matches πr^2 for a circle of radius 2 (since $r = 4\sin\theta$ traces a circle of radius 2 centered at $(0, 2)$): $\pi(2)^2 = 4\pi$ ✓.

17 [Challenging] • Target: Comparing polar area to rectangular area (conceptual) • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Explain why the polar area formula $A = \frac{1}{2}\int r^2 \, d\theta$ uses r^2 rather than just r, by relating it to the area of a thin circular sector.

A thin sector of angle $d\theta$ and radius r has area approximately $\frac{1}{2}r^2 \, d\theta$ (a fraction $\frac{d\theta}{2\pi}$ of the full circle's area πr^2). Summing (integrating) these thin sector areas over the range of θ gives the total polar area formula — the r^2 arises naturally from the geometry of a circular sector, not from any arbitrary convention.

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18 [Challenging] • Target: Comparing polar area to rectangular area (conceptual) • Cross-Lesson: 3.01 Parametric Equations

A student claims the area between $r = 1$ and $r = 2$ (an annulus, full circle) is simply $\frac{1}{2}\int_0^{2\pi} (2-1)^2\, d\theta$. Explain the error and give the correct setup.

The error is subtracting the radii before squaring — the correct polar formula requires subtracting the SQUARED radii, not squaring the difference of radii. Correct setup: $A = \frac{1}{2}\int_0^{2\pi} (2^2 - 1^2) d\theta = \frac{1}{2}\int_0^{2\pi} 3\, d\theta = 3\pi$ (matching $\pi(2^2 - 1^2) = 3\pi$ from the standard annulus-area formula).

19 [Challenging] • Target: Area between two polar curves • Cross-Lesson: 3.02 Vector-Valued Functions

Convert the rectangular point $(0, -3)$ to polar coordinates.

Theorem (Rectangular to Polar Conversion): For rectangular coordinates (x, y), the polar radius and angle satisfy: $r = \sqrt{x^2 + y^2}$, $\tan\theta = \frac{y}{x}$ (with quadrant resolution). Step 1 · Compute radius r: $r = \sqrt{0^2 + (-3)^2} = \sqrt{0 + 9} = \sqrt{9} = 3$. Step 2 · Determine angle θ: The point $(0, -3)$ lies on the negative y-axis. Moving counterclockwise from the positive x-axis, the directional angle is $\theta = \frac{3\pi}{2}$ (or $-\frac{\pi}{2}$). Conclusion: In standard polar coordinates with $r > 0$: $\mathbf{\left(3, \frac{3\pi}{2}\right)}$ or equivalently $\mathbf{\left(3, -\frac{\pi}{2}\right)}$.

20 [Challenging] • Target: Area of one petal of a rose curve • Cross-Lesson: 4.01 Polar Curves and Derivatives

Convert the polar point $(2, \frac{\pi}{2})$ to rectangular coordinates.

Theorem (Polar to Rectangular Conversion): For a point specified in polar coordinates (r, θ), its Cartesian rectangular coordinates (x, y) are: $x = r\cos\theta$, $y = r\sin\theta$. Step 1 · Identify coordinates: Radius: $r = 2$; Angle: $\theta = \frac{\pi}{2}$. Step 2 · Compute x and y: $x = 2\cos\left(\frac{\pi}{2}\right) = 2(0) = 0$, $y = 2\sin\left(\frac{\pi}{2}\right) = 2(1) = 2$. Conclusion: The rectangular coordinates are: **(0, 2)**.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The polar area formula

Find the area enclosed by $r = 1$ for $\theta \in [0, \frac{\pi}{2}]$.

a. $\frac{\pi}{4}$ ✓

b. $\frac{\pi}{2}$

c. $-\frac{7\pi}{4}$

d. $\frac{3\pi}{4}$

$A = \frac{1}{2}\int_0^{\frac{\pi}{2}} 1^2\, d\theta = \frac{1}{2}(1)(\frac{\pi}{2}) = \frac{\pi}{4}$.

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22 [Foundational] • Target: Area of a full cardioid

Find the area enclosed by $r = 3$ for $\theta \in [0, \frac{3\pi}{2}]$.

a. $\frac{31\pi}{4}$

b. $\frac{27\pi}{2}$

c. $\frac{27\pi}{4}$ ✓

d. $\frac{29\pi}{4}$

$A = \frac{1}{2}\int_0^{\frac{3\pi}{2}} 3^2\, d\theta = \frac{1}{2}(9)(\frac{3\pi}{2}) = \frac{27\pi}{4}$.

23 [Foundational] • Target: Area between two polar curves

Find the area enclosed by $r = 1$ for $\theta \in [0, \pi]$.

a. $-\frac{3\pi}{2}$

b. π

c. $\frac{5\pi}{2}$

d. $\frac{\pi}{2}$ ✓

$A = \frac{1}{2}\int_0^{\pi} 1^2\, d\theta = \frac{1}{2}(1)(\pi) = \frac{\pi}{2}$.

24 [Foundational] • Target: Area of one petal of a rose curve

Find the area enclosed by $r = 6$ for $\theta \in [0, \frac{\pi}{2}]$.

a. 18π

b. 7π

c. $\frac{19\pi}{2}$

d. 9π ✓

$A = \frac{1}{2}\int_0^{\frac{\pi}{2}} 6^2\, d\theta = \frac{1}{2}(36)(\frac{\pi}{2}) = 9\pi$.

25 [Foundational] • Target: Setting up area between curves that intersect

Find the area enclosed by $r = 9$ for $\theta \in [0, \frac{\pi}{2}]$.

a. $\frac{81\pi}{4}$ ✓

b. $\frac{81\pi}{2}$

c. $\frac{89\pi}{4}$

d. $\frac{83\pi}{4}$

$A = \frac{1}{2}\int_0^{\frac{\pi}{2}} 9^2\, d\theta = \frac{1}{2}(81)(\frac{\pi}{2}) = \frac{81\pi}{4}$.

26 [Foundational] • Target: Area of a full rose curve

Find the area enclosed by $r = 9$ for $\theta \in [0, \pi]$.

a. 81π

b. $\frac{83\pi}{2}$

c. $\frac{81\pi}{2}$ ✓

d. 41π

$A = \frac{1}{2}\int_0^{\pi} 9^2\, d\theta = \frac{1}{2}(81)(\pi) = \frac{81\pi}{2}$.

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27 [Foundational] • Target: Area inside a limaçon

Find the area enclosed by $r = 8$ for $\theta \in [0, 2\pi]$.

a. $\frac{129\pi}{2}$

b. 62π

c. 64π ✓

d. 128π

$A = \frac{1}{2}\int_0^{2\pi} 8^2\, d\theta = \frac{1}{2}(64)(2\pi) = 64\pi.$

28 [Foundational] • Target: Setting up area problems (AP-style, no evaluation)

Find the area inside the circle $r = 9$ but outside the circle $r = 4$.

a. -7π

b. 25π

c. 77π

d. 65π ✓

$A = \frac{1}{2}\int_0^{2\pi} (9^2 - 4^2)d\theta = \frac{1}{2}(65)(2\pi) = 65\pi.$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Finding intersection points of polar curves

Find the area inside the circle $r = 8$ but outside the circle $r = 5$.

a. 39π ✓

b. 59π

c. 9π

d. -17π

$A = \frac{1}{2}\int_0^{2\pi} (8^2 - 5^2)d\theta = \frac{1}{2}(39)(2\pi) = 39\pi.$

30 [Intermediate] • Target: Area using symmetry to simplify computation

Find the area inside the circle $r = 13$ but outside the circle $r = 6$.

a. 163π

b. -23π

c. 49π

d. 133π ✓

$A = \frac{1}{2}\int_0^{2\pi} (13^2 - 6^2)d\theta = \frac{1}{2}(133)(2\pi) = 133\pi.$

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31 [Intermediate] • Target: Area review: full worked example

Find the area inside the circle $r = 10$ but outside the circle $r = 5$.

a. -15π

b. 75π ✓

c. 95π

d. 25π

$A = \frac{1}{2}\int_0^{2\pi}(10^2 - 5^2)d\theta = \frac{1}{2}(75)(2\pi) = 75\pi.$

32 [Intermediate] • Target: Comparing polar area to rectangular area (conceptual)

Find the area inside the circle $r = 8$ but outside the circle $r = 3$.

a. 55π ✓

b. 61π

c. 25π

d. $-\pi$

$A = \frac{1}{2}\int_0^{2\pi}(8^2 - 3^2)d\theta = \frac{1}{2}(55)(2\pi) = 55\pi.$

33 [Intermediate] • Target: Sector Area Formula & Single Curves

Find the area inside the circle $r = 10$ but outside the circle $r = 3$.

a. 49π

b. 97π

c. 91π ✓

d. π

$A = \frac{1}{2}\int_0^{2\pi}(10^2 - 3^2)d\theta = \frac{1}{2}(91)(2\pi) = 91\pi.$

34 [Intermediate] • Target: Area Between Two Polar Curves

Find the total area enclosed by the curve $r = 8 + 7\sin\theta$.

a. 113π

b. $\frac{177\pi}{2}$ ✓

c. $\frac{179\pi}{2}$

d. $\frac{225\pi}{2}$

Expanding $(8 + 7\sin\theta)^2 = 8^2 + 2(8)(7)\sin\theta + 7^2\sin^2\theta$ and integrating over $[0, 2\pi]$, the linear $\sin\theta$ term vanishes and $\sin^2\theta$ averages to $\frac{1}{2}$, leaving $A = \pi\left(8^2 + \frac{7^2}{2}\right) = \frac{177\pi}{2}$.

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35 [Intermediate] • Target: Common Interior of Two Curves

Find the total area enclosed by the curve $r = 5 + 9\sin\theta$.

a. 98π

b. 106π

c. $\frac{133\pi}{2}$

d. $\frac{131\pi}{2}$ ✓

Expanding $(5 + 9\sin\theta)^2 = 5^2 + 2(5)(9)\sin\theta + 9^2\sin^2\theta$ and integrating over $[0, 2\pi]$, the linear $\sin\theta$ term vanishes and $\sin^2\theta$ averages to $\frac{1}{2}$, leaving $A = \pi\left(5^2 + \frac{9^2}{2}\right) = \frac{131\pi}{2}$.

36 [Intermediate] • Target: Integration Limits & Full-Period Curves

Find the total area enclosed by the curve $r = 3 + 5\cos\theta$.

a. 34π

b. 32π

c. $\frac{45\pi}{2}$

d. $\frac{43\pi}{2}$ ✓

Expanding $(3 + 5\cos\theta)^2 = 3^2 + 2(3)(5)\cos\theta + 5^2\cos^2\theta$ and integrating over $[0, 2\pi]$, the linear $\cos\theta$ term vanishes and $\cos^2\theta$ averages to $\frac{1}{2}$, leaving $A = \pi\left(3^2 + \frac{5^2}{2}\right) = \frac{43\pi}{2}$.

37 [Intermediate] • Target: The polar area formula

Find the area of one petal of $r = 4\sin(7\theta)$.

a. $\frac{23\pi}{28}$

b. $\frac{8\pi}{7}$

c. $\frac{16\pi}{7}$

d. $\frac{4\pi}{7}$ ✓

Using $A_{petal} = \frac{k^2\pi}{4n}$ with $k = 4$, $n = 7$: $A = \frac{16\pi}{28} = \frac{4\pi}{7}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Area of a full cardioid • Cross-Lesson: 3.01 Parametric Equations

Find the area of one petal of $r = 4\sin(2\theta)$.

a. $\frac{5\pi}{2}$

b. 8π

c. 4π

d. 2π ✓

Using $A_{petal} = \frac{k^2\pi}{4n}$ with $k = 4$, $n = 2$: $A = \frac{16\pi}{8} = 2\pi$.

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39 [Challenging] • Target: Area between two polar curves • Cross-Lesson: 3.02 Vector-Valued Functions

Find the area inside the circle $r = 11$ but outside the circle $r = 7$, for $\theta \in [0, \pi]$ only (half the full region).

a. 8π

b. 72π

c. 36π ✓

d. 37π

$A = \frac{1}{2}\int_0^{\pi}(11^2 - 7^2)d\theta = \frac{1}{2}(72)(\pi) = 36\pi.$

40 [Challenging] • Target: Area of one petal of a rose curve • Cross-Lesson: 4.01 Polar Curves and Derivatives

Find the area inside the circle $r = 3$ but outside the circle $r = 1$, for $\theta \in [0, \pi]$ only (half the full region).

a. 5π

b. 8π

c. 4π ✓

d. 2π

$A = \frac{1}{2}\int_0^{\pi}(3^2 - 1^2)d\theta = \frac{1}{2}(8)(\pi) = 4\pi.$

41 [Challenging] • Target: Sector Area Formula & Single Curves • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Which integral formula calculates the area of a region bounded by the polar curve $r = f(\theta)$ from α to β?

a. $\frac{1}{2}\int_{\alpha}^{\beta} r^2 d\theta$ ✓

b. $\int_{\alpha}^{\beta} r^2 d\theta$

c. $\frac{1}{2}\int_{\alpha}^{\beta} r d\theta$

d. $\pi\int_{\alpha}^{\beta} r^2 d\theta$

Polar area is the sum of infinitesimal circular sectors with area $dA = \frac{1}{2}r^2 d\theta$.

42 [Challenging] • Target: Area Between Two Polar Curves • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

Which integral represents the area inside $r = R(\theta)$ and outside $r = r(\theta)$?

a. $\frac{1}{2}\int_{\alpha}^{\beta}(R(\theta)^2 - r(\theta)^2)d\theta$ ✓

b. $\frac{1}{2}\int_{\alpha}^{\beta}(R(\theta) - r(\theta))^2 d\theta$

c. $\int_{\alpha}^{\beta}(R^2 - r^2)d\theta$

d. $\frac{1}{2}\int_{\alpha}^{\beta} R(\theta)r(\theta)d\theta$

Area is difference of sectors: $\frac{1}{2}\int(R^2 - r^2)d\theta$.

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43 [Challenging] • Target: Area Between Two Polar Curves • Cross-Lesson: 6.01 Power Series and Interval of Convergence

Set up the integral for the area inside $r = 6\cos\theta$ and outside $r = 3$.

a. $\frac{1}{2}\int_{-\pi/3}^{\pi/3}(36\cos^2\theta - 9)d\theta$ ✓

b. $\frac{1}{2}\int_{-\pi/3}^{\pi/3}(6\cos\theta - 3)^2 d\theta$

c. $\int_{0}^{\pi/3}(36\cos^2\theta - 9)d\theta$

d. $\frac{1}{2}\int_{0}^{\pi/3}(6\cos\theta - 3)d\theta$

Outer curve is $6\cos\theta$, inner curve is 3. Formula: $\frac{1}{2}\int_{-\pi/3}^{\pi/3}(36\cos^2\theta - 9)d\theta$.

44 [Challenging] • Target: Common Interior of Two Curves • Cross-Lesson: 6.02 Taylor and Maclaurin Series

Find the area inside both $r = 2$ and $r = 4\sin\theta$ in the first quadrant.

a. $\frac{4\pi}{3} - \sqrt{3}$ ✓

b. 2π

c. $\frac{2\pi}{3}$

d. $4\sqrt{3}$

Intersection: $4\sin\theta = 2 \Rightarrow \sin\theta = 1/2 \Rightarrow \theta = \pi/6$. From 0 to $\pi/6$, bound is $4\sin\theta$; from $\pi/6$ to $\pi/2$, bound is 2. Sum yields $\frac{4\pi}{3} - \sqrt{3}$.

45 [Challenging] • Target: Integration Limits & Full-Period Curves • Cross-Lesson: 1.01 Integration by Parts

What is the range of θ required to trace the limaçon $r = 2 + \cos\theta$ exactly once?

a. $[0, 2\pi]$ ✓

b. $[0, \pi]$

c. $[0, 4\pi]$

d. $[-\frac{\pi}{2}, \frac{\pi}{2}]$

For a limaçon without loops ($r = a + b\cos\theta$ with $a > b$), the curve is traced exactly once as θ traverses $[0, 2\pi]$.

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